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1 Heat Equation

Take the 1D heat equation

$$u_t - \nu u_{xx} = 0, \quad \nu > 0$$

with initial condition $u(x, 0) = f(x)$ and 2π -periodic boundary conditions.

It is perhaps the most well-known partial differential equation. As its name suggests, it is designed to model how heat diffuses in a spatial region over time. In fluid dynamics, it can be used to model how the concentration of a pollutant diffuses in a fluid body under random Brownian motion. Qualitatively, it is well known that it exhibits a field-smoothing behaviour, evening out rough and jagged parts of the solution. Despite describing these interesting and physical phenomena, the heat equation is also highly approachable, both analytically and numerically. It is therefore a good place to start. We will solve it analytically using two different methods. The first will be the standard variable separation method. The second will be a Fourier method. We will do this for a couple of reasons. Firstly, there seems to be a relative lack of approachable literature on the heat equation with periodic boundary conditions compared to Dirichlet or Neumann boundary conditions. Also, we will be able to see the comparative brevity of the Fourier method which will eventually inspire a numerical algorithm that we will be able to use for other partial differential equations.

1.1 Method of Separation of Variables

This is a standard method for finding an exact solution to the heat equation. We start by imposing a variable separability condition on our solution and see what has to follow.

Separable solution Ansatz:

$$u(x, t) = X(x)T(t).$$

Substituting into the heat equation, it follows that

$$XT' - \nu X''T = 0.$$

Rewriting the equation:

$$\frac{1}{\nu} \frac{T'}{T} = \frac{X''}{X}.$$

We notice how both sides of the equality are independent of each other's variable, and yet equality holds for all domain values of x and t . This is only possible if both sides are constant:

$$\frac{1}{\nu} \frac{T'}{T} = \lambda = \frac{X''}{X}.$$

1.1.1 Spatial Component

We begin by solving the spatial equation, which is just a second order homogeneous ordinary differential equation:

$$X'' - \lambda X = 0.$$

which gives us the spatial component

$$X(x) = Ae^{i\sqrt{\lambda}x} + Be^{-i\sqrt{\lambda}x},$$

for arbitrary constants A and B .

Case 1: $\lambda > 0$, i.e. $\lambda = \mu^2$ for some $\mu \in \mathbb{R} \setminus \{0\}$.

Then we have

$$X'' - \mu^2 X = 0,$$

which has general solution

$$X(x) = Ae^{-\mu x} + Be^{\mu x}.$$

Applying periodicity, we require that $X(0) = X(2\pi)$ and $X'(0) = X'(2\pi)$, which gives

$$A + B = Ae^{-2\pi\mu} + Be^{2\pi\mu}$$

and

$$A - B = Ae^{-2\pi\mu} - Be^{2\pi\mu}.$$

Combining, we can see that

$$A = Ae^{-2\pi\mu}, \quad B = Be^{2\pi\mu}.$$

This means either $A = B = 0$ or $\mu = 0$. but we had imposed $\mu \neq 0$, so we obtain only the trivial solution $X(x) = 0$.

Case 2: $\lambda = 0$.

In this case, the spatial equation immediately simplifies to

$$X''(x) = 0,$$

giving us the general solution

$$X''(x) = Ax + B.$$

Our boundary conditions give

$$A + Bx = A + 2\pi Bx,$$

which forces $B = 0$, giving us another trivial solution $X(x) = A$.

Case 3: $\lambda < 0$, i.e. $\lambda = -k^2$ for some $k \in \mathbb{R} \setminus \{0\}$.

The equation becomes

$$X'' + k^2 X = 0,$$

with general solution

$$X(x) = Ae^{ikx} + Be^{-ikx}.$$

Enforcing periodicity:

$$X(0) = X(2\pi) \Rightarrow A + B = Ae^{2\pi ikx} + Be^{-2\pi ikx}.$$

Rearranging, we get

$$A(1 - e^{2\pi ikx}) + B(1 - e^{-2\pi ikx}) = 0.$$

We also need $X'(0) = X'(2\pi)$, which gives

$$Aik - Bik = Aike^{2\pi ikx} - Bike^{-2\pi ikx},$$

which simplifies to the following

$$A(1 - e^{2\pi ikx}) - B(1 - e^{-2\pi ikx}) = 0.$$

Combining these two conditions, we can state that

$$2A(1 - e^{2\pi ikx}) = 0,$$

So either $A = 0$, or $e^{2\pi ikx} = 1, \forall x \in \mathbb{R}$.

Similarly,

$$2B(1 - e^{-2\pi ikx}) = 0,$$

So either $B = 0$ or $e^{-2\pi ikx} = 1, \forall x \in \mathbb{R}$.

For a nontrivial solution, we choose either $A \neq 0$ or $B \neq 0$. Regardless, we obtain the constraint $k \in \mathbb{Z}$.

1.1.2 Temporal Component

For the temporal component we can see

$$T' = \nu\lambda T.$$

Not forgetting that we found the constraint $\lambda = -k^2, k \in \mathbb{Z}$, we have the equation

$$T' = -\nu k^2 T,$$

which we quickly solve as

$$T(t) = T(0)e^{-\nu k^2 t}.$$

1.1.3 General Solution

Putting the spatial and temporal components together, we obtain a solution

$$u_k(x, t) = T_k(0)e^{-\nu k^2 t}(A_k e^{ikx} + B_k e^{-ikx}), \quad k \in \mathbb{Z}.$$

Since the heat equation is linear, we make use of the superposition principle. To arrive at our general solution, we sum over $k \in \mathbb{Z}$. In doing so, we can absorb constants A_k , B_k , and $T_k(0)$ into a single constant C_k .

$$u(x, t) = \sum_{k=-\infty}^{\infty} \hat{f}_k e^{-\nu k^2 t} e^{ikx},$$

where

$$\hat{f}_k = \langle u(x, 0), e^{ikx} \rangle = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-ikx} dx.$$

1.2 Fourier Method

We approach the same problem set-up. We seek a piecewise continuous function in $L^2([0, 2\pi])$, so we pre-emptively write the solution as

$$u(x, t) = \sum_{k=-\infty}^{\infty} \hat{u}_k(t) e^{ikx},$$

where

$$\hat{u}_k(t) = \langle u(x, t), e^{ikx} \rangle = \frac{1}{2\pi} \int_0^{2\pi} u(x, t) e^{-ikx} dx.$$

We simply insert this form into the heat equation

$$\begin{aligned} \frac{\partial}{\partial t} \left(\sum_{k=-\infty}^{\infty} \hat{u}_k e^{ikx} \right) - \frac{\partial^2}{\partial x^2} \left(\nu \sum_{k=-\infty}^{\infty} \hat{u}_k e^{ikx} \right) &= 0 \\ \Rightarrow \sum_{k=-\infty}^{\infty} \frac{d\hat{u}_k}{dt} e^{ikx} - \nu \sum_{k=-\infty}^{\infty} \hat{u}_k \frac{d^2}{dx^2} e^{ikx} &= 0 \\ \Rightarrow \sum_{k=-\infty}^{\infty} \frac{d\hat{u}_k}{dt} e^{ikx} - \nu \sum_{k=-\infty}^{\infty} (ik)^2 \hat{u}_k e^{ikx} &= 0 \\ \Rightarrow \sum_{k=-\infty}^{\infty} \frac{d\hat{u}_k}{dt} e^{ikx} = \sum_{k=-\infty}^{\infty} -\nu k^2 \hat{u}_k e^{ikx}. \end{aligned}$$

Since this is an expansion in orthonormal basis functions e^{ikx} , we can equate coefficients to obtain a family of ODEs

$$\frac{d\hat{u}_k}{dt} = -\nu k^2 \hat{u}_k,$$

which resolve easily as

$$\hat{u}_k(t) = \hat{u}_k(0)e^{-\nu k^2 t} = \hat{f}_k e^{-\nu k^2 t}.$$

Thus we obtain our solution

$$u(x, t) = \sum_{k=-\infty}^{\infty} \hat{f}_k e^{-\nu k^2 t} e^{ikx}.$$

We can see how much more direct the Fourier solution method is compared to the classic separation of variables, as it allows us to reduce the heat equation into a family of simple ordinary differential equations. And regardless of method, we already see the utility of Fourier analysis by looking at the solution expression, where we can see the intuitively understood field-smoothing actually instantiated analytically by the decay term $e^{-\nu k^2 t}$ – we see that over time, for larger and larger Fourier modes, k , we have smaller and smaller contributions to the solution.